
Math 22

Geometry

Math 22 Course at a Glance

Geometry

Review Content

REVIEW	
M22-REV-001	I can solve a linear equation arising from a geometric relationship.
M22-REV-002	I can solve a proportion and interpret the resulting scale factor.
M22-REV-003	I can simplify an exact square root and approximate it when needed.
M22-REV-004	I can calculate distance, midpoint, and slope on the coordinate plane.
M22-REV-005	I can substitute measurements into a formula with consistent units.

Geometry Basics

REQUIRED	
M22-BAS-001	I can name and represent points, lines, planes, segments, and rays using correct notation.
M22-BAS-002	I can identify collinear, coplanar, intersecting, and concurrent figures.
M22-BAS-003	I can calculate segment lengths using betweenness and the segment-addition postulate.
M22-BAS-004	I can determine a midpoint or use a segment bisector relationship.
M22-BAS-005	I can classify and measure angles.
M22-BAS-006	I can calculate angle measures using angle addition and angle bisectors.
M22-BAS-007	I can identify and use complementary, supplementary, vertical, and linear-pair relationships.
M22-BAS-008	I can distinguish a definition, postulate, theorem, converse, and counterexample.
M22-BAS-009	I can determine angle relationships formed by parallel lines and a transversal.
M22-BAS-010	I can use angle relationships to determine whether lines are parallel.
M22-BAS-011	I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.
M22-BAS-012	I can construct a logical geometric argument from stated facts and accepted results.

Triangle Congruence and Proof

REQUIRED	
M22-CON-001	I can classify triangles by sides and angles.
M22-CON-002	I can use triangle angle-sum and exterior-angle relationships.
M22-CON-003	I can write a congruence statement with corresponding vertices in the correct order.
M22-CON-004	I can determine whether triangles are congruent by SAS.
M22-CON-005	I can determine whether triangles are congruent by SSS.
M22-CON-006	I can determine whether right triangles are congruent by HL.
M22-CON-007	I can determine whether triangles are congruent by ASA or AAS.
M22-CON-008	I can apply isosceles and equilateral triangle theorems and their converses.
M22-CON-009	I can use corresponding parts of congruent triangles after proving congruence.
M22-CON-010	I can write a structured proof involving triangle congruence.

Quadrilaterals, Perimeter, and Area

REQUIRED	
M22-QUAD-001	I can calculate interior or exterior angle measures of a polygon.
M22-QUAD-002	I can calculate perimeter and area of common polygons.
M22-QUAD-003	I can apply properties of parallelograms to determine unknown measures.
M22-QUAD-004	I can determine whether a quadrilateral is a parallelogram.
M22-QUAD-005	I can apply properties of rectangles, rhombi, and squares.
M22-QUAD-006	I can classify a quadrilateral from its stated or marked properties.
M22-QUAD-007	I can apply trapezoid and trapezoid-midsegment relationships.
M22-QUAD-008	I can apply properties of kites.
M22-QUAD-009	I can calculate the perimeter or area of a composite polygon.
M22-QUAD-010	I can justify a quadrilateral conclusion using definitions and theorems.

Math 22 Course at a Glance

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Similarity and Proportional Reasoning

REQUIRED	
M22-SIM-001	I can determine image lengths and coordinates under a dilation.
M22-SIM-002	I can identify the scale factor of a dilation or similarity relationship.
M22-SIM-003	I can write a similarity statement with corresponding vertices in the correct order.
M22-SIM-004	I can use proportional corresponding sides to solve similar-polygon problems.
M22-SIM-005	I can relate similarity scale factor to perimeter and area scale factors.
M22-SIM-006	I can prove triangles similar by AA, SSS, or SAS.
M22-SIM-007	I can apply triangle proportionality theorems and their converses.
M22-SIM-008	I can solve and justify an indirect-measurement or other similarity application.

Right Triangles

REQUIRED	
M22-RGT-001	I can apply the Pythagorean theorem to determine a missing side.
M22-RGT-002	I can use the converse of the Pythagorean theorem to classify a triangle.
M22-RGT-003	I can apply triangle-inequality relationships.
M22-RGT-004	I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.
M22-RGT-005	I can apply right-triangle similarity relationships.
M22-RGT-006	I can use geometric-mean relationships created by an altitude to the hypotenuse.
M22-RGT-007	I can select sine, cosine, or tangent for a right-triangle problem.
M22-RGT-008	I can determine a missing side or angle of a right triangle.
M22-RGT-009	I can solve a contextual right-triangle problem.
M22-RGT-010	I can justify a right-triangle solution using an appropriate theorem or ratio.

Circles

REQUIRED	
M22-CIR-001	I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.
M22-CIR-002	I can calculate circumference and arc length.
M22-CIR-003	I can calculate the area of a circle, sector, or segment.
M22-CIR-004	I can determine angle and arc measures involving central and inscribed angles.
M22-CIR-005	I can apply relationships among chords, arcs, and distances from the center.
M22-CIR-006	I can apply tangent-radius and tangent-segment relationships.
M22-CIR-007	I can determine angle measures formed by chords, secants, or tangents.
M22-CIR-008	I can determine segment lengths formed by intersecting chords.
M22-CIR-009	I can determine segment lengths formed by secants or tangents.
M22-CIR-010	I can solve a multi-step circle problem and justify each relationship used.

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

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Review Content

REVIEW

M22-REV-001

I can solve a linear equation arising from a geometric relationship.

SUPPORTING SKILLS

A	Check a proposed solution in the original equation. (first introduced in Math 12) SK-ALG-CHECK-SOLUTION · Apply
R	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Reinforce
A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply

M22-REV-002

I can solve a proportion and interpret the resulting scale factor.

SUPPORTING SKILLS

R	Use cross multiplication when two ratios form a proportion. (inherited from middle school) SK-ALG-CROSS-MULTIPLY · Reinforce
I	Calculate and interpret a similarity or dilation scale factor. SK-SIM-SCALE-FACTOR · Introduce

M22-REV-003

I can simplify an exact square root and approximate it when needed.

SUPPORTING SKILLS

I	Distinguish exact values from measured or rounded approximations. SK-GEO-EXACT-APPROX · Introduce
R	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Reinforce

M22-REV-004

I can calculate distance, midpoint, and slope on the coordinate plane.

SUPPORTING SKILLS

R	Calculate horizontal or vertical distance from the difference of coordinates. (inherited horizontal/vertical coordinate distance) SK-COORD-DISTANCE-AXIS · Reinforce
I	Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. SK-COORD-DISTANCE-PYTHAGOREAN · Introduce
R	Calculate the midpoint of a segment from endpoint coordinates. (inherited from earlier mathematics) SK-COORD-MIDPOINT · Reinforce
R	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Reinforce

M22-REV-005

I can substitute measurements into a formula with consistent units.

SUPPORTING SKILLS

I	Select and report appropriate linear, square, or angular units. SK-GEO-UNITS · Introduce
A	Apply the order of operations to a numerical expression. (inherited from middle school) SK-NUM-ORDER-OPERATIONS · Apply

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Geometry Basics

REQUIRED

M22-BAS-001	I can name and represent points, lines, planes, segments, and rays using correct notation.
SUPPORTING SKILLS	
I	Add congruence, parallel, perpendicular, and angle markings to a diagram. SK-GEO-DIAGRAM-MARK · Introduce
I	Read and write geometric notation correctly. SK-GEO-NOTATION · Introduce

M22-BAS-002	I can identify collinear, coplanar, intersecting, and concurrent figures.
SUPPORTING SKILLS	
I	Distinguish stated or marked facts from visual appearance. SK-GEO-DIAGRAM-INFER · Introduce
A	Read and write geometric notation correctly. (first introduced in Math 22) SK-GEO-NOTATION · Apply

M22-BAS-003	I can calculate segment lengths using betweenness and the segment-addition postulate.
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
I	Translate segment addition into an equation. SK-GEO-SEGMENT-ADDITION · Introduce

M22-BAS-004	I can determine a midpoint or use a segment bisector relationship.
SUPPORTING SKILLS	
I	Use a bisector to create equal measures. SK-GEO-BISECTOR · Introduce
I	Use a midpoint to create equal segment lengths. SK-GEO-MIDPOINT · Introduce
A	Translate segment addition into an equation. (first introduced in Math 22) SK-GEO-SEGMENT-ADDITION · Apply

M22-BAS-005	I can classify and measure angles.
SUPPORTING SKILLS	
I	Identify complementary, supplementary, vertical, and linear-pair relationships. SK-ANG-PAIR-IDENTIFY · Introduce
I	Measure or calculate an angle accurately. SK-GEO-ANGLE-MEASURE · Introduce

M22-BAS-006	I can calculate angle measures using angle addition and angle bisectors.
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Translate angle addition into an equation. SK-GEO-ANGLE-ADDITION · Introduce
A	Use a bisector to create equal measures. (first introduced in Math 22) SK-GEO-BISECTOR · Apply

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M22-BAS-007

I can identify and use complementary, supplementary, vertical, and linear-pair relationships.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
D	Identify complementary, supplementary, vertical, and linear-pair relationships. (first introduced in Math 22) SK-ANG-PAIR-IDENTIFY · Deepen

M22-BAS-008

I can distinguish a definition, postulate, theorem, converse, and counterexample.

SUPPORTING SKILLS

I	Recognize and apply the converse of a conditional statement. SK-PROOF-CONVERSE · Introduce
I	Disprove a universal statement with a counterexample. SK-PROOF-COUNTEREXAMPLE · Introduce
I	Apply a geometric definition as justification. SK-PROOF-DEFINITION · Introduce
I	Select a theorem that connects known facts to a target statement. SK-PROOF-THEOREM-SELECT · Introduce

M22-BAS-009

I can determine angle relationships formed by parallel lines and a transversal.

SUPPORTING SKILLS

I	Use parallel-line angle relationships. SK-ANG-PARALLEL-APPLY · Introduce
I	Identify corresponding, alternate, and same-side angle pairs. SK-ANG-TRANSVERSAL-IDENTIFY · Introduce

M22-BAS-010

I can use angle relationships to determine whether lines are parallel.

SUPPORTING SKILLS

I	Use angle relationships to prove lines parallel. SK-ANG-PARALLEL-CONVERSE · Introduce
A	Identify corresponding, alternate, and same-side angle pairs. (first introduced in Math 22) SK-ANG-TRANSVERSAL-IDENTIFY · Apply

M22-BAS-011

I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Use parallel-line angle relationships. (first introduced in Math 22) SK-ANG-PARALLEL-APPLY · Apply
I	Use right-angle relationships created by perpendicular lines. SK-LIN-PERPENDICULAR-ANGLE · Introduce

M22-BAS-012

I can construct a logical geometric argument from stated facts and accepted results.

SUPPORTING SKILLS

I	Translate givens into diagram markings and statements. SK-PROOF-GIVEN · Introduce
I	Organize statements and reasons into a valid proof sequence. SK-PROOF-STRUCTURE · Introduce
A	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Apply

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Triangle Congruence and Proof

REQUIRED

M22-CON-001	I can classify triangles by sides and angles.
SUPPORTING SKILLS	
A	Add congruence, parallel, perpendicular, and angle markings to a diagram. (first introduced in Math 22) SK-GEO-DIAGRAM-MARK · Apply
I	Classify a triangle by side lengths and angle measures. SK-TRI-CLASSIFY · Introduce

M22-CON-002	I can use triangle angle-sum and exterior-angle relationships.
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Apply the triangle angle-sum theorem. SK-TRI-ANGLE-SUM · Introduce
I	Apply the triangle exterior-angle theorem. SK-TRI-EXTERIOR-ANGLE · Introduce

M22-CON-003	I can write a congruence statement with corresponding vertices in the correct order.
SUPPORTING SKILLS	
I	Match corresponding vertices, sides, and angles in order. SK-GEO-CORRESPONDENCE · Introduce
I	Write congruence statements in corresponding order. SK-PROOF-CONGRUENCE-ORDER · Introduce

M22-CON-004	I can determine whether triangles are congruent by SAS.
SUPPORTING SKILLS	
A	Match corresponding vertices, sides, and angles in order. (first introduced in Math 22) SK-GEO-CORRESPONDENCE · Apply
D	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Deepen
I	Select SAS, SSS, HL, ASA, or AAS from marked information. SK-TRI-CONGRUENCE-SELECT · Introduce

M22-CON-005	I can determine whether triangles are congruent by SSS.
SUPPORTING SKILLS	
A	Match corresponding vertices, sides, and angles in order. (first introduced in Math 22) SK-GEO-CORRESPONDENCE · Apply
D	Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22) SK-TRI-CONGRUENCE-SELECT · Deepen

M22-CON-006	I can determine whether right triangles are congruent by HL.
SUPPORTING SKILLS	
A	Use right-angle relationships created by perpendicular lines. (first introduced in Math 22) SK-LIN-PERPENDICULAR-ANGLE · Apply
D	Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22) SK-TRI-CONGRUENCE-SELECT · Deepen

M22-CON-007	I can determine whether triangles are congruent by ASA or AAS.
SUPPORTING SKILLS	
A	Apply the triangle angle-sum theorem. (first introduced in Math 22) SK-TRI-ANGLE-SUM · Apply
D	Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22) SK-TRI-CONGRUENCE-SELECT · Deepen

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M22-CON-008	I can apply isosceles and equilateral triangle theorems and their converses.
SUPPORTING SKILLS	
D	Recognize and apply the converse of a conditional statement. (first introduced in Math 22) SK-PROOF-CONVERSE · Deepen
I	Apply isosceles-triangle theorems and converses. SK-TRI-ISOSCELES · Introduce

M22-CON-009	I can use corresponding parts of congruent triangles after proving congruence.
SUPPORTING SKILLS	
A	Write congruence statements in corresponding order. (first introduced in Math 22) SK-PROOF-CONGRUENCE-ORDER · Apply
I	Use corresponding parts only after triangle congruence is established. SK-PROOF-CPCTC · Introduce

M22-CON-010	I can write a structured proof involving triangle congruence.
SUPPORTING SKILLS	
A	Use corresponding parts only after triangle congruence is established. (first introduced in Math 22) SK-PROOF-CPCTC · Apply
A	Translate givens into diagram markings and statements. (first introduced in Math 22) SK-PROOF-GIVEN · Apply
D	Organize statements and reasons into a valid proof sequence. (first introduced in Math 22) SK-PROOF-STRUCTURE · Deepen
A	Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22) SK-TRI-CONGRUENCE-SELECT · Apply

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Quadrilaterals, Perimeter, and Area

REQUIRED

M22-QUA D-001 I can calculate interior or exterior angle measures of a polygon.

SUPPORTING SKILLS

A

Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)
SK-ALG-INVERSE-OPERATIONS · Apply

I

Calculate polygon interior or exterior angle sums.
SK-POLY-ANGLE-SUM · Introduce

M22-QUA D-002 I can calculate perimeter and area of common polygons.

SUPPORTING SKILLS

I

Add boundary lengths without omitting or duplicating sides.
SK-GEO-PERIMETER · Introduce

D

Select and report appropriate linear, square, or angular units. (first introduced in Math 22)
SK-GEO-UNITS · Deepen

A

Apply the order of operations to a numerical expression. (inherited from middle school)
SK-NUM-ORDER-OPERATIONS · Apply

M22-QUA D-003 I can apply properties of parallelograms to determine unknown measures.

SUPPORTING SKILLS

A

Translate quantities and relationships into an equation. (first introduced in Math 12)
SK-ALG-TRANSLATE-EQUATION · Apply

I

Apply parallelogram properties and tests.
SK-QUAD-PARALLELOGRAM · Introduce

M22-QUA D-004 I can determine whether a quadrilateral is a parallelogram.

SUPPORTING SKILLS

A

Recognize and apply the converse of a conditional statement. (first introduced in Math 22)
SK-PROOF-CONVERSE · Apply

D

Apply parallelogram properties and tests. (first introduced in Math 22)
SK-QUAD-PARALLELOGRAM · Deepen

M22-QUA D-005 I can apply properties of rectangles, rhombi, and squares.

SUPPORTING SKILLS

I

Use the inclusive hierarchy of quadrilateral classifications.
SK-QUAD-HIERARCHY · Introduce

I

Distinguish and apply rectangle, rhombus, and square properties.
SK-QUAD-RECT-RHOM-SQUARE · Introduce

M22-QUA D-006 I can classify a quadrilateral from its stated or marked properties.

SUPPORTING SKILLS

A

Distinguish stated or marked facts from visual appearance. (first introduced in Math 22)
SK-GEO-DIAGRAM-INFER · Apply

D

Use the inclusive hierarchy of quadrilateral classifications. (first introduced in Math 22)
SK-QUAD-HIERARCHY · Deepen

M22-QUA D-007 I can apply trapezoid and trapezoid-midsegment relationships.

SUPPORTING SKILLS

A

Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)
SK-ALG-INVERSE-OPERATIONS · Apply

I

Apply trapezoid and midsegment relationships.
SK-QUAD-TRAPEZOID · Introduce

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M22-QUA D-008	I can apply properties of kites.
SUPPORTING SKILLS	
I	Apply kite properties. SK-QUAD-KITE · Introduce

M22-QUA D-009	I can calculate the perimeter or area of a composite polygon.
SUPPORTING SKILLS	
I	Decompose a composite region into familiar shapes. SK-GEO-AREA-DECOMPOSE · Introduce
D	Add boundary lengths without omitting or duplicating sides. (first introduced in Math 22) SK-GEO-PERIMETER · Deepen
A	Select and report appropriate linear, square, or angular units. (first introduced in Math 22) SK-GEO-UNITS · Apply

M22-QUA D-010	I can justify a quadrilateral conclusion using definitions and theorems.
SUPPORTING SKILLS	
D	Apply a geometric definition as justification. (first introduced in Math 22) SK-PROOF-DEFINITION · Deepen
A	Organize statements and reasons into a valid proof sequence. (first introduced in Math 22) SK-PROOF-STRUCTURE · Apply
D	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Deepen

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Similarity and Proportional Reasoning

REQUIRED

M22-SIM-001	I can determine image lengths and coordinates under a dilation.
SUPPORTING SKILLS	
A	Match corresponding vertices, sides, and angles in order. (first introduced in Math 22) SK-GEO-CORRESPONDENCE · Apply
D	Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22) SK-SIM-SCALE-FACTOR · Deepen

M22-SIM-002	I can identify the scale factor of a dilation or similarity relationship.
SUPPORTING SKILLS	
A	Use cross multiplication when two ratios form a proportion. (inherited from middle school) SK-ALG-CROSS-MULTIPLY · Apply
D	Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22) SK-SIM-SCALE-FACTOR · Deepen

M22-SIM-003	I can write a similarity statement with corresponding vertices in the correct order.
SUPPORTING SKILLS	
D	Match corresponding vertices, sides, and angles in order. (first introduced in Math 22) SK-GEO-CORRESPONDENCE · Deepen
I	Write similarity statements in corresponding order. SK-PROOF-SIMILARITY-ORDER · Introduce

M22-SIM-004	I can use proportional corresponding sides to solve similar-polygon problems.
SUPPORTING SKILLS	
A	Use cross multiplication when two ratios form a proportion. (inherited from middle school) SK-ALG-CROSS-MULTIPLY · Apply
A	Match corresponding vertices, sides, and angles in order. (first introduced in Math 22) SK-GEO-CORRESPONDENCE · Apply
I	Set up proportions using corresponding sides. SK-SIM-PROPORTION · Introduce

M22-SIM-005	I can relate similarity scale factor to perimeter and area scale factors.
SUPPORTING SKILLS	
I	Relate length scale factor to perimeter and area factors. SK-SIM-PERIMETER-AREA · Introduce
A	Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22) SK-SIM-SCALE-FACTOR · Apply

M22-SIM-006	I can prove triangles similar by AA, SSS, or SAS.
SUPPORTING SKILLS	
A	Write similarity statements in corresponding order. (first introduced in Math 22) SK-PROOF-SIMILARITY-ORDER · Apply
D	Organize statements and reasons into a valid proof sequence. (first introduced in Math 22) SK-PROOF-STRUCTURE · Deepen
I	Recognize AA triangle similarity. SK-SIM-AA · Introduce
I	Test SSS or SAS triangle similarity. SK-SIM-SSS-SAS · Introduce

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M22-SIM-007	I can apply triangle proportionality theorems and their converses.
SUPPORTING SKILLS	
A	Recognize and apply the converse of a conditional statement. (first introduced in Math 22) SK-PROOF-CONVERSE · Apply
I	Apply triangle proportionality relationships. SK-SIM-TRI-PROPORTIONALITY · Introduce

M22-SIM-008	I can solve and justify an indirect-measurement or other similarity application.
SUPPORTING SKILLS	
A	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Apply
D	Set up proportions using corresponding sides. (first introduced in Math 22) SK-SIM-PROPORTION · Deepen
A	Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22) SK-SIM-SCALE-FACTOR · Apply

Right Triangles

REQUIRED

M22-RGT-001	I can apply the Pythagorean theorem to determine a missing side.
SUPPORTING SKILLS	
A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply
I	Apply the Pythagorean theorem or its converse. SK-TRI-PYTHAGOREAN · Introduce

M22-RGT-002	I can use the converse of the Pythagorean theorem to classify a triangle.
SUPPORTING SKILLS	
A	Recognize and apply the converse of a conditional statement. (first introduced in Math 22) SK-PROOF-CONVERSE · Apply
D	Apply the Pythagorean theorem or its converse. (first introduced in Math 22) SK-TRI-PYTHAGOREAN · Deepen

M22-RGT-003	I can apply triangle-inequality relationships.
SUPPORTING SKILLS	
A	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Apply
A	Classify a triangle by side lengths and angle measures. (first introduced in Math 22) SK-TRI-CLASSIFY · Apply

M22-RGT-004	I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.
SUPPORTING SKILLS	
A	Distinguish exact values from measured or rounded approximations. (first introduced in Math 22) SK-GEO-EXACT-APPROX · Apply
R	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Reinforce
I	Recall special-right-triangle ratios. SK-TRI-SPECIAL-RATIOS · Introduce

M22-RGT-005	I can apply right-triangle similarity relationships.
SUPPORTING SKILLS	
A	Recognize AA triangle similarity. (first introduced in Math 22) SK-SIM-AA · Apply
D	Set up proportions using corresponding sides. (first introduced in Math 22) SK-SIM-PROPORTION · Deepen

M22-RGT-006	I can use geometric-mean relationships created by an altitude to the hypotenuse.
SUPPORTING SKILLS	
A	Set up proportions using corresponding sides. (first introduced in Math 22) SK-SIM-PROPORTION · Apply
I	Apply geometric-mean relationships in a right triangle split by an altitude. SK-TRI-GEOMETRIC-MEAN · Introduce

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M22-RGT-007

I can select sine, cosine, or tangent for a right-triangle problem.

SUPPORTING SKILLS

A	Add congruence, parallel, perpendicular, and angle markings to a diagram. (first introduced in Math 22) SK-GEO-DIAGRAM-MARK · Apply
I	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. SK-TRI-TRIG-RATIO · Introduce
I	Label sides relative to a selected angle. SK-TRIG-TRIANGLE-LABEL · Introduce

M22-RGT-008

I can determine a missing side or angle of a right triangle.

SUPPORTING SKILLS

A	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Apply
I	Use an inverse trigonometric function to determine a missing angle. SK-TRI-TRIG-SOLVE-ANGLE · Introduce
I	Use a trigonometric ratio to determine a missing side. SK-TRI-TRIG-SOLVE-SIDE · Introduce

M22-RGT-009

I can solve a contextual right-triangle problem.

SUPPORTING SKILLS

A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
D	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Deepen
A	Use an inverse trigonometric function to determine a missing angle. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-ANGLE · Apply
A	Use a trigonometric ratio to determine a missing side. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-SIDE · Apply

M22-RGT-010

I can justify a right-triangle solution using an appropriate theorem or ratio.

SUPPORTING SKILLS

D	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Deepen
A	Apply the Pythagorean theorem or its converse. (first introduced in Math 22) SK-TRI-PYTHAGOREAN · Apply
A	Recall special-right-triangle ratios. (first introduced in Math 22) SK-TRI-SPECIAL-RATIOS · Apply
A	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Apply

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Circles

REQUIRED

M22-CIR-001

I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.

SUPPORTING SKILLS

I	Identify circle parts and relationships from a diagram. SK-CIR-VOCAB · Introduce
A	Read and write geometric notation correctly. (first introduced in Math 22) SK-GEO-NOTATION · Apply

M22-CIR-002

I can calculate circumference and arc length.

SUPPORTING SKILLS

I	Relate a central angle to a fraction of a circle. SK-CIR-ARC-FRACTION · Introduce
I	Calculate arc length from radius and central angle. SK-CIR-ARC-LENGTH · Introduce
I	Calculate circumference from radius or diameter. SK-CIR-CIRCUMFERENCE · Introduce
A	Select and report appropriate linear, square, or angular units. (first introduced in Math 22) SK-GEO-UNITS · Apply

M22-CIR-003

I can calculate the area of a circle, sector, or segment.

SUPPORTING SKILLS

I	Calculate the area of a circle. SK-CIR-AREA · Introduce
I	Calculate sector area from radius and central angle. SK-CIR-SECTOR-AREA · Introduce
I	Calculate segment area by subtracting a triangle from a sector. SK-CIR-SEGMENT-AREA · Introduce
A	Select and report appropriate linear, square, or angular units. (first introduced in Math 22) SK-GEO-UNITS · Apply

M22-CIR-004

I can determine angle and arc measures involving central and inscribed angles.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
D	Relate a central angle to a fraction of a circle. (first introduced in Math 22) SK-CIR-ARC-FRACTION · Deepen
I	Relate an inscribed angle to its intercepted arc. SK-CIR-INSCRIBED-ANGLE · Introduce

M22-CIR-005

I can apply relationships among chords, arcs, and distances from the center.

SUPPORTING SKILLS

I	Apply equal-chord, equal-arc, and center-distance relationships. SK-CIR-CHORD-RELATION · Introduce
A	Identify circle parts and relationships from a diagram. (first introduced in Math 22) SK-CIR-VOCAB · Apply

M22-CIR-006

I can apply tangent-radius and tangent-segment relationships.

SUPPORTING SKILLS

I	Select and apply an intersecting-segment product relationship. SK-CIR-SEGMENT-PRODUCT · Introduce
I	Use the perpendicular relationship between a radius and tangent. SK-CIR-TANGENT-RADIUS · Introduce

M22-CIR-007

I can determine angle measures formed by chords, secants, or tangents.

SUPPORTING SKILLS

I	Select the correct angle formula for chords, secants, or tangents. SK-CIR-ANGLE-RELATION · Introduce
A	Identify circle parts and relationships from a diagram. (first introduced in Math 22) SK-CIR-VOCAB · Apply

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M22-CIR-008

I can determine segment lengths formed by intersecting chords.

SUPPORTING SKILLS

A

Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)
SK-ALG-INVERSE-OPERATIONS · Apply

I

Select and apply an intersecting-segment product relationship.
SK-CIR-SEGMENT-PRODUCT · Introduce

M22-CIR-009

I can determine segment lengths formed by secants or tangents.

SUPPORTING SKILLS

A

Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)
SK-ALG-INVERSE-OPERATIONS · Apply

D

Select and apply an intersecting-segment product relationship. (first introduced in Math 22)
SK-CIR-SEGMENT-PRODUCT · Deepen

M22-CIR-010

I can solve a multi-step circle problem and justify each relationship used.

SUPPORTING SKILLS

A

Select the correct angle formula for chords, secants, or tangents. (first introduced in Math 22)
SK-CIR-ANGLE-RELATION · Apply

A

Select and apply an intersecting-segment product relationship. (first introduced in Math 22)
SK-CIR-SEGMENT-PRODUCT · Apply

A

Identify circle parts and relationships from a diagram. (first introduced in Math 22)
SK-CIR-VOCAB · Apply

D

Select a theorem that connects known facts to a target statement. (first introduced in Math 22)
SK-PROOF-THEOREM-SELECT · Deepen